

On the origin of scientific peer effects, strategic complementarity or conformity?

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Abstract

Collaboration plays a critical role in scientific productivity, yet the mechanisms through which peer effects drive innovation remain underexplored. This paper investigates the microfoundations of scientific peer effects, focusing on how collaboration influences individual productivity. While existing literature emphasizes the importance of peer effects, it has not fully addressed the underlying processes. This work contributes by offering a clearer understanding of these mechanisms and providing insights for innovation policy, particularly in contexts where collaboration drives knowledge creation. The results indicate that peer effects stem from both strategic complementarity and conformity: collaborating with more productive coauthors boosts individual performance, while deviating from group norms reduces utility. These findings underscore the significant role of network externalities in shaping scientific outcomes and the structure of scientific collaboration.

Keywords: productivity, innovation, network, peer effects, knowledge production, scientific collaboration

JEL: C72, D85, D43, L14, Z13

1. Introduction

Technological advancement is a key driver of economic growth, but the mechanisms underlying knowledge creation, which is essential for innovation, remain complex and not fully understood. Endogenous growth models propose that new knowledge arises from recombining existing knowledge stocks (Romer, 1990; Weitzman, 1998), yet these models do not fully capture the individual and social dynamics involved. Recent research indicates that variation in scientists' attributes, such as curiosity, experience, and domain expertise, significantly influences research productivity (Fleming, 2001; Fortunato et al., 2018). At the same time, knowledge creation has become increasingly collaborative (Wuchty et al., 2007), as researchers share ideas and access a wider knowledge base, which enhances scientific output. Empirical studies employing quasi-natural experiments highlight the causal importance of peer interactions and coauthorship networks in advancing scientific progress (Azoulay et al., 2010; Waldinger, 2012).

Social networks research has shown that an agent’s position within a network correlates with their outcomes. This relationship is often analyzed using centrality measures, which quantify an agent’s importance based on their network connections. Benlahlou (2024) demonstrate that Katz-Bonacich centrality outperforms other centrality measures in capturing the link between network position and scientific productivity. However, Katz-Bonacich centrality can be interpreted through two distinct microfoundations: strategic complementarity and social conformity (Ushchev and Zenou, 2020). While both mechanisms are represented in the same Nash Katz-Bonacich linkage, they lead to distinct policy implications. Therefore, disentangling these mechanisms is crucial for understanding their roles in scientific productivity and for developing more effective network-based policies that promote knowledge production.

Understanding the magnitude and sources of individual-level spillovers is crucial for innovation policy. The impact of scientists’ actions extends beyond their own behavior, generating broader ”multiplier effects” that shape the overall production process. By exploring these microfoundations, I aim to deepen our understanding of how scientists’ interactions and decisions influence scientific productivity. By combining a micro-founded model with a robust empirical strategy, I am able to identify causal relationships that enhance our theoretical understanding of scientific peer-effects and inform more effective policies to promote scientific progress and economic growth.

The microfoundations of this model build on the framework developed by Liu et al. (2014), which integrates two key models of peer effects in networks: the local-aggregate and local-average models. In the local-aggregate model (Ballester et al., 2006; Belhaj et al., 2016; Calvó-Armengol et al., 2009; König et al., 2014), peer effects arise from the total output of coauthors. Here, the productivity of a scientist’s peers directly increases their marginal utility from producing science. On the other hand, the local-average model (Patacchini and Zenou, 2012; Boucher et al., 2014) attributes peer effects to conformity. In this model, scientists incur a cost for deviating from the average production level of their coauthors, leading them to adjust their output to align with the social norm defined by their peers’ average productivity.

I test these models using a dataset focused on cancer-related research, a field where innovation is often driven by scientific collaboration rather than isolated efforts. This emphasis is particularly relevant given the substantial economic burden of cancer. According to the National Institutes of Health, cancer care cost the U.S. \$147.5 billion in 2015, while lost productivity from premature cancer-related

deaths accounted for an additional \$134.8 billion in 2005. The dataset combines detailed information extracted from CVs with disambiguated publication records from the PubMed Knowledge Graph (Xu et al., 2020), supplemented by metadata from the Torvik Research Group.

I adopt an instrumental variable (IV) strategy to estimate the best-response function implied by the theoretical model, allowing the identification of peer effects. To address the potential endogeneity of collaboration networks, I construct instruments using predicted coauthorship ties based on predetermined dyadic characteristics (Liu et al., 2014). Specifically, in the first stage, I model the probability of a coauthorship link using early-career overlap in research interests and observable attributes such as gender, ethnicity, and institutional affiliation. In the second stage, the predicted network is used to instrument the observed coauthorship structure, allowing for causal identification of peer effects within the collaborative network. The underlying idea is that early-career similarity in research interests has a persistent influence on the likelihood of collaboration later in one’s career (Ductor, 2015).

I find that peer effects arise from both strategic complementarity and conformity simultaneously, even though estimating each model in isolation may lead to misleading conclusions. These findings illustrate the potential significance and magnitude of network externalities in scientific productivity and provide insights into the mechanisms underlying peer effects. A thorough understanding of these origins is crucial for designing more effective public policies that leverage the amplifying power of networks.

This paper contributes to various strands of the literature. First, it adds to research on the impact of intellectual collaboration on research output. Existing studies show that co-authorship improves publication quality and quantity (Hollis, 2001), and that early collaboration networks predict future success (Ductor et al., 2014). Productivity also declines following the loss of collaborators, reflecting knowledge spillovers (Azoulay et al., 2010; Waldinger, 2012). On the other hand, the rise of research teams is well documented and linked to increasing knowledge complexity (Wuchty et al., 2007; Jones et al., 2008; Azoulay et al., 2019; Jaravel et al., 2018; Jones, 2009). I extend this literature by identifying two microfoundations, strategic complementarity and conformity, that explain the productivity gains from collaboration.

Additionally, from an empirical perspective, this analysis intersects with the extensive literature on peer effects in economics (Bramoullé et al., 2020), which builds on the seminal work of Manski (1993). I extend the estimator developed by Liu et al. (2014) to account for network endogeneity, drawing on methods from Lee et al. (2021) and Kelejian and Piras (2014). To my knowledge, this is the first study

to model peer effects as jointly driven by strategic complementarity and conformity, while accommodating both network endogeneity and heteroscedasticity.

The remainder of the paper is organized as follows. Section 1 presents a game-theoretical framework of scientific peer effects that forms the basis for the subsequent empirical analysis. In Section 2, I discuss the identification threats and outline the methodology used to estimate the model. Section 3 describes the data used in the analysis. Section 4 presents the results from a structural estimation to estimate peer effects among scientists, allowing for heterogeneity with respect to network position. Section 5 concludes.

2. Theoretical framework

The central empirical question addressed in this study is whether peer effects in scientific production arise from strategic complementarities, social conformity, or a combination of both. To test this hypothesis, I build upon the peer effect model of Liu et al. (2014). This model captures both strategic complementarity and conformism, while also encompassing two standard models of peer effects: the local-average and local-aggregate models.

The model rests on two key assumptions: first, that scientists benefit from the increased production of their peers; and second, that they incur a cost when deviating from the norms of their reference group. In equilibrium, a scientist's production is a function of their network position, with their incentives shaped by the choices of coauthors through both strategic complementarity and conformism, as well as by their own individual scientific productivity.

2.1. The Network and Notations

Suppose a finite set of scientists $N = \{1, \dots, n\}$ is divided into communities, where $N_c = \{1, \dots, n_c\}$ represents the scientists in the c -th community. Coauthorship connections within community c are tracked through its adjacency matrix $G_c = [g_{ij,c}]$, where $g_{ij,c} = g_{ji,c} = 1$ if i and j are coauthors, and $g_{ij,c} = 0$ otherwise. Additionally, $g_{ii,c} = 0$. The reference group of scientist i in community c consists of i 's coauthors, denoted as $N_{i,c} = \{j \neq i \mid g_{ij,c} = 1\}$. The size of $N_{i,c}$, termed the degree of i in graph theory, is $|N_{i,c}| = |\{j \neq i \mid g_{ij,c} = 1\}|$. Let $G_c^* = [g_{ij,c}^*]$, where $g_{ij,c}^* = g_{ij,c}/g_{i,c}$.

I denote by $y_{i,c}$ the production level of researcher i and by $\mathbf{y}_c$ the population production profile of community c .

2.2. Modeling Benefits of Collaboration

The literature on peer effects in scientific productivity implicitly acknowledges the concept of strategic complementarity, whereby the actions of peers positively influence a scientist’s productivity and the quality of their research. Such effects operate through various channels, including the exchange of insights, the provision of feedback, and the collaborative development of methodologies. Collaborative efforts, such as co-authorship, combine diverse expertise and foster an environment rich in intellectual exchange.

Working with innovative and productive peers often grants access to valuable ideas, especially when strong connections are maintained. Early exposure to such ideas significantly enhances scientific productivity, indicating that researchers closely associated with highly productive peers tend to achieve higher output themselves. Consequently, an individual’s productivity is shaped not only by their own characteristics but also by the productivity of their collaborators, reflecting its recursive nature and dependence on peer interactions.

The literature on scientific peer effects highlights two key factors that influence the benefits of collaboration: strategic complementarity and heterogeneity in idiosyncratic productivity. To capture these elements, I model the benefit from production as a function of individual effort, idiosyncratic ability, and the output of one’s co-authors:

$$\underbrace{\left(\pi_{i,c}^* + \lambda_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} \right)}_{\text{benefits}} y_{i,c}$$

In this model, $y_{i,c}$ represents the production level of scientist i , and the term $\left(\pi_{i,c}^* + \lambda_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} \right)$ represents the return to production. Here, $\pi_{i,c}^*$ captures individual heterogeneity in production returns, and $\sum_{j \in N_{i,c}} y_{j,c}$ represents the aggregate production of i ’s co-authors. The social multiplier coefficient, $\lambda_1 \geq 0$, captures the extent to which peer production influences individual utility. Importantly, this model allows for heterogeneity in the network structure, as the aggregate production of co-authors differs for each scientist, even if all co-authors choose the same production level.

2.3. Modeling the Cost of Collaboration

In the sociology of science, recognition plays a crucial role in shaping scientists’ behaviors and motivations. A substantial body of literature emphasizes that scientists actively seek validation and recognition from their peers, considering such acknowledgment a key incentive in academic work. This

recognition serves as an affirmation of the quality and significance of their work (Zuckerman, 1977; Latour and Woolgar, 1986). However, the pursuit of recognition and validation within collaborative environments entails various costs for individual scientists.

In team-based research, recognizing individual contributions becomes increasingly complex as the boundaries between personal and collective achievements blur. The subjective nature of recognition and validation within scientific communities adds another layer of difficulty to the evaluation process. Researchers often hold differing views on the significance and impact of their work, which can create challenges in collaborative settings. In this context, reassurance through recognition plays a vital role in the social dynamics of knowledge production. Despite their expertise, scientists frequently face uncertainty about the value and influence of their contributions. As a result, peer recognition serves as a crucial form of validation, affirming the importance of their work within the broader scientific community (Merton, 1973).

According to Festinger (1954), individuals tend to compare themselves to others with similar attitudes or abilities. Peer groups act as benchmarks or reference points that individuals use to assess situations and their own standing within them (Mas and Moretti, 2009). These groups provide a realistic benchmark for assessing one's position and achievements within a broader social context. The desire to relate to and be accepted by a group drives ambition and competitive behavior, as individuals seek alignment with their peers (Shibutani, 1955).

To account for the costs incurred by a researcher, I follow the standard approach in economics to model conformity, i.e., using an increasing function of the distance to the average output of peers (Akerlof, 1997). However, it is interesting to note that, in my case, the social norm is not exogenously determined. The model also includes a quadratic term to account for increasing marginal costs.

$$\underbrace{\frac{1}{2} \left[y_{i,c}^2 + \lambda_2 \left(y_{i,c} - \sum_{j=1}^{n_c} g_{ij,c}^* y_{j,c} \right)^2 \right]}_{\text{cost}}$$

Here, $\lambda_2 \geq 0$ captures the intensity of the conformity effect.

2.4. The Peer Effects Game

Given the underlying network structure represented by the adjacency matrix G_c , individuals in community c simultaneously decide how much science they produce $y_{i,c}$ to maximize the following utility function:

$$\begin{aligned}
u_{i,c}(y_{i,c}) \equiv u_{i,c}(y_{i,c}, \mathbf{Y}_c, \mathbf{G}_c) = & \underbrace{\left(\pi_{i,c}^* + \lambda_1 \sum_{j=1}^n g_{ij,c} y_{j,c} \right)}_{\text{benefits}} y_{i,c} \\
& - \underbrace{\frac{1}{2} \left[y_{i,c}^2 + \lambda_2 (y_{i,c} - \sum_{j=1}^n g_{ij,c}^* y_{j,c})^2 \right]}_{\text{cost}}.
\end{aligned} \tag{1}$$

Now, I will point out some important properties of the utility function (1), which provide useful intuition about the mechanism at work. First, when scientist j produces, they exert a negative externality on their coauthor i if and only if i 's production deviates from their social norm. Second, productions are strategic complements, meaning that the higher the production of a scientist's coauthor, the higher the individual's marginal utility of producing science. Hence, a scientist's utility increases with the aggregate production of their coauthors but decreases as their own output deviates from the average production of those coauthors. Deriving from the first-order condition of utility maximization, the best-reply function of author i is then expressed as:

$$y_{i,c} = \phi_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} + \phi_2 \sum_{j=1}^{n_c} g_{ij,c}^* y_{j,c} + \pi_{i,c} \tag{2}$$

where $\phi_1 = \frac{\lambda_1}{1+\lambda_2}$, $\phi_2 = \frac{\lambda_2}{1+\lambda_2}$, and $\pi_{i,c} = \frac{\pi_{i,c}^*}{1+\lambda_2}$. As $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$, I have $\phi_1 \geq 0$ and $0 \leq \phi_2 < 1$. The coefficient ϕ_1 is called the local-aggregate endogenous peer effect. As $\phi_1 \geq 0$, this coefficient reflects strategic complementarity in efforts. The coefficient ϕ_2 is called the local-average endogenous peer effect, which captures the taste for conformity. Note that $\frac{\phi_1}{\phi_2} = \frac{\lambda_1}{\lambda_2}$. That is, the relative magnitude of ϕ_1 and ϕ_2 is the same that of the strategic complementarity coefficient λ_1 and the social-conformity coefficient λ_2 .

I denote by $g_c^{\max}$ the highest degree in network c , i.e. $g_c^{\max} = \max_i g_{i,c}$. Let $\Pi_c = (\pi_{1,c}), \dots, \pi_{n_c,c}$. The Nash equilibrium of the general network model is characterized by the following proposition:

Proposition 1 (Liu et al., 2014)

If $\phi_1 \geq 0$, $\phi_2 \geq 0$, and $g_c^{\max} \phi_1 + \phi_2 < 1$, then the network game with payoffs (1) has a unique interior Nash equilibrium in pure strategies given by:

$$\mathbf{Y}_c = (\mathbf{I}_{n_c} - \phi_1 \mathbf{G}_c - \phi_2 \mathbf{G}_c^*)^{-1} \Pi_c$$

I am dealing with a game with strategic complementarities, so the higher the production of coauthors, the higher my marginal utility of raising my own production. Consequently, there is a problem of equilibrium existence since there is no bound on the production of each agent. Since ϕ_1 and ϕ_2 express the intensity of these network effects, the condition $g_c^{\max}\phi_1 + \phi_2 < 1$ guarantees the existence of equilibrium by bounding the relative strength of strategic complementarities of production with respect to the amplifying power of the network. The linearity of best-reply functions and strategic complementarities interact to impose mutual discipline, thereby ruling out the possibility of multiple equilibria.

2.5. Model Discussion

In the pure conformity case (i.e., the local-average model with $\lambda_1 = 0$), a scientist's utility depends on the extent to which their own production deviates from the average production of their coauthors. Therefore, the closer a researcher's output is to the average output of their coauthors, the higher their equilibrium utility. In the case of pure strategic complementarities, heterogeneity in network position gives rise to variation in equilibrium output levels, even when researchers are ex ante identical in terms of their individual productivity. In the local-average model, network positions do not affect equilibrium effort if all scientists are ex ante identical. However, in the pure strategic complementarity scenario, different network positions result in different equilibrium production levels, even if scientists are ex ante identical in terms of individual productivity.

These two mechanisms have fundamentally different equilibrium implications. When a scientist's utility is influenced by the aggregate production (strategic complementarities) of coauthors, their position in the network affects their equilibrium production. Conversely, when a deviation from the social norm is costly, all scientists will conform to the production level of their reference group, resulting in uniform production if they are ex ante identical.

Strategic complementarities and conformity can each generate the same reduced form, Katz-Bonacich linkage to equilibrium production, making it difficult to distinguish between the two.

However, this distinction implies that the policy implications of the two mechanisms differ significantly. In the case of conformity, influencing scientists' choices and outcomes requires shifting the prevailing social norm within the community. As a result, policies must influence the majority of the scientific community to be effective. Group-based interventions such as topic-based initiatives are therefore particularly well-suited to this context. Conversely, with strategic complementarities, due

to social multiplier effects, targeting a single scientist can still yield positive outcomes, as that scientist will influence their coauthors. Hence, scientist-based policies, such as targeted subsidies, can be effectively implemented in this scenario.

3. Identification of the econometric network model

The following discussion builds on the foundational work of Liu et al. (2014), extending their analysis to address potential endogenous network and heteroscedasticity.

3.1. Econometric network model

The specification of the econometric model follows the equilibrium best-reply function of the network game so that it has a clear microfoundation. Let the ex ante heterogeneity $\pi_{i,c}$ of scientist i in community c be

$$\pi_{i,c} = \eta_c + x_{i,c}^\top \beta_1 + u_{i,c},$$

where $x_{i,c}$ is a p -dimensional vector of exogenous variables, η_c a constant term, β_1 is a vector of suitable dimension, where the error term u is assumed to be independently distributed but allowed to be heteroscedastic with $\Sigma \equiv \mathbb{E}(\mathbf{u}\mathbf{u}^\top) = \sigma_i^2$ being a diagonal matrix, and β_1 are corresponding parameters. From the best-reply function (2), the general econometric network model is

$$y_{i,c} = \phi_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} + \phi_2 \sum_{j=1}^{n_c} g_{ij,c}^* y_{j,c} + \eta_c + x_{i,c}^\top \beta_1 + u_{i,c}, \quad (3)$$

for $i = 1, \dots, n_c$, and $c = 1, \dots, \bar{c}$. Let $\mathbf{Y}_c = (y_{1,c}, \dots, y_{n_c,c})^\top$, $\mathbf{X}_c = (x_{1,c}, \dots, x_{n_c,c})^\top$, and $\mathbf{u}_c = (u_{1,c}, \dots, u_{n_c,c})^\top$. Then (3) can be written in matrix form as

$$\mathbf{Y}_c = \phi_1 \mathbf{G}_c \mathbf{Y}_c + \phi_2 \mathbf{G}_c^* \mathbf{Y}_c + \eta_c \mathbf{1}_{n_c} + \mathbf{X}_c \beta_1 + \mathbf{u}_c.$$

Let $\text{diag}(A_{ij})$ denote a block diagonal matrix in which the diagonal blocks are $m_j \times n_j$ matrices A_{js} . For a data set with $\bar{c}$ communities, let $\mathbf{Y} = (\mathbf{Y}'_1, \dots, \mathbf{Y}'_{\bar{c}})'$, $\mathbf{X} = (\mathbf{X}'_1, \dots, \mathbf{X}'_{\bar{c}})'$, $\mathbf{u} = (\mathbf{u}'_1, \dots, \mathbf{u}'_{\bar{c}})'$, $\eta = (\eta'_1, \dots, \eta'_{\bar{c}})'$, $\mathbf{G} = \{G_c\}_{c=1}^{c=\bar{c}}$, $\mathbf{G}^* = \{G_c^*\}_{c=1}^{c=\bar{c}}$ and $\mathbf{L} = \{\mathbf{1}_{n,c}\}_{c=1}^{c=\bar{c}}$. The econometric model can be written as

$$\mathbf{Y} = \phi_1 \mathbf{G} \mathbf{Y} + \phi_2 \mathbf{G}^* \mathbf{Y} + \mathbf{X} \beta_1 + \mathbf{L} \eta + \mathbf{u}$$

The network-specific parameters, denoted by η , are allowed to vary with $\mathbf{G}$, $\mathbf{G}^*$, and $\mathbf{X}$, similar to the structure in a fixed-effects panel data model. To mitigate the incidental parameter problem that

arises when the number of groups is large, I eliminate the term $\mathbf{L}\eta$ by applying the deviation from the group mean projection, where $\mathbf{J}_c = \mathbf{I}_{n_c} - \frac{1}{n_c}\iota_{n_c}\iota_{n_c}'$. This transformation is conceptually similar to the within-transformation used in fixed-effects panel data models. As $\mathbf{J}\mathbf{L} = 0$, the transformed network model is

$$\mathbf{J}\mathbf{Y} = \phi_1\mathbf{J}\mathbf{G}\mathbf{Y} + \phi_2\mathbf{J}\mathbf{G}^*\mathbf{Y} + \mathbf{J}\mathbf{X}_c\beta_1 + \mathbf{J}\mathbf{L}\eta + \mathbf{J}\mathbf{u} \quad (4)$$

The econometric network model (4) incorporates the *endogenous effect*, captured by ϕ_1 and ϕ_2 , where a scientist's choice may depend on those of their coauthors and the effect of idiosyncratic characteristics is captured by β_1 . Furthermore, I distinguish between the *aggregate endogenous effect*, captured by the coefficient ϕ_1 , and the *average endogenous effect*, captured by the coefficient ϕ_2 , as they originate from different economic models with totally different equilibrium implications.

3.2. Estimation strategy

My goal is to consistently estimate the parameters $\theta = (\phi_1, \phi_2, \beta_1')^\top$ in Equation (4) considering the potential threats to identification exposed in the following.

3.2.1. Threats to identification

Manski (1993) identifies two principal sources of peer influence relevant to this study. Endogenous effects occur when an individual's behavior is directly shaped by the actions of their peers. In contrast, correlated effects arise from shared unobserved factors or environments that lead individuals to behave similarly in the absence of direct peer influence. Distinguishing between these effects is critical in order to obtain causal estimates from equation (4).

A major challenge lies in the endogenous formation of coauthorship networks, where scientists strategically select collaborators, resulting in reciprocal influence and confounding. Moreover, highly productive researchers tend to cluster non-randomly, further biasing estimates. In addition, mechanical associations between individuals' outcomes and community characteristics can produce spurious correlations.

To address these challenges, the coauthorship formation process can be explicitly modeled using dyadic regression (Graham and De Paula, 2020). This approach accounts for the fact that collaborations are not formed randomly but reflect strategic decisions based on expected benefits. The resulting predicted network structure can then be used to construct instrumental variables that, when combined with fixed effects, improve identification by isolating peer influence from confounding factors.

3.2.2. Identification challenges and instrumental variables

When the adjacency matrix $\mathbf{G}$ is exogenous, linear social interaction models in the form of Equation (4) can be estimated by the 2SLS based on the linear moment condition $\mathbf{Z}^\top u(\theta) = 0$, where $\mathbf{Z}$ is a matrix of IVs consisting of linearly independent columns of $\mathbf{J}[\mathbf{X}, \mathbf{G}^*\mathbf{X}, \mathbf{GX}, \mathbf{G}^{*2}\mathbf{X}]$, let also $\underline{\mathbf{X}} = \mathbf{J}[\mathbf{GY}, \mathbf{G}^*\mathbf{Y}, \mathbf{X}, \mathbf{G}^*\mathbf{X}]$ and

$$\mathbf{Ju}(\theta) = \mathbf{JY} - \underline{\mathbf{X}}\theta$$

The parameters in Equation (4) can be identified, provided that a certain level of intransitivity exists in the network such that the usual rank condition of IV estimators holds (Bramoullé et al., 2009; Liu and Lee, 2010; Liu et al., 2014).

If an unobserved individual factor influences both research output and coauthorship connections, the adjacency matrix $\mathbf{G}$ becomes endogenous, which invalidates the instrumental variables matrix $\mathbf{Z}$. To address this issue, a valid instrument matrix can be obtained by replacing the observed adjacency matrix $\mathbf{G}$ with a predicted matrix $\hat{\mathbf{G}}$, constructed from exogenous covariates (Kelejian and Piras, 2014).

In particular, for each pair of individuals i and j within a community, the utility that individual i derives from forming a link with individual j can be expressed as:

$$U_{ij} = \delta_0 + W_{ij}\delta_1 + \varepsilon_{ij},$$

where U_{ij} represents the utility scientist i obtains from coauthoring with scientist j , W_{ij} is the deterministic component of utility, specified as a function of observable characteristics of both i and j , and ε_{ij} captures unobserved factors influencing link formation. Assuming that ε_{ij} is independently and identically distributed following a Type I extreme value distribution, the resulting model for dyadic network formation model corresponds to a logistic regression.

$$\mathbb{P}(g_{ij} = 1) = \frac{\exp(\delta_0 + W_{ij}\delta_1)}{1 + \exp(\delta_0 + W_{ij}\delta_1)}. \quad (5)$$

Based on Equation (5), define

$$\hat{g}_{ij} = \frac{\exp(\hat{\delta}_0 + W_{ij}\hat{\delta}_1)}{1 + \exp(\hat{\delta}_0 + W_{ij}\hat{\delta}_1)}$$

where $\hat{\delta}_0$ and $\hat{\delta}_1$ are convergent estimate. To ensure that the predicted adjacency matrix is uniformly bounded in row and column sums, I normalize $\hat{g}_{ij}$ by dividing it by $\hat{d} = \max \left(\max_i \sum_{j=1}^n \hat{g}_{ij}, \max_j \sum_{i=1}^n \hat{g}_{ij} \right)$ (Kelejian and Prucha, 2010), and define the (i, j) th element of the predicted adjacency matrix $\hat{G}$ as $\hat{g}_{ij}/\hat{d}$ if $i \neq j$, and zero otherwise.

The IV matrix based on the predicted adjacency matrix $\hat{\mathbf{G}}$ is denoted by $\hat{\mathbf{Z}}$ and includes linearly independent columns of $\mathbf{J}[\mathbf{X}, \bar{\mathbf{G}}\mathbf{X}, \hat{\mathbf{G}}\mathbf{X}, \hat{\mathbf{G}}\bar{\mathbf{G}}\mathbf{X}]$, the corresponding 2SLS estimator is given by

$$\hat{\theta}_{2sls} = \left[\underline{\mathbf{X}}^\top \hat{\mathbf{Z}} \left(\hat{\mathbf{Z}}^\top \hat{\mathbf{Z}} \right)^{-1} \hat{\mathbf{Z}}^\top \underline{\mathbf{X}} \right]^{-1} \underline{\mathbf{X}}^\top \hat{\mathbf{Z}} \left(\hat{\mathbf{Z}}^\top \hat{\mathbf{Z}} \right)^{-1} \hat{\mathbf{Z}}^\top \mathbf{J} \mathbf{y}. \quad (6)$$

As pointed by Lee et al. (2021), The consistency of the proposed 2SLS estimator does not rely on the consistency of the estimator $\hat{\delta} = \left(\hat{\delta}_0, \hat{\delta}_1^\top \right)^\top$.

4. Network and Data

This section is divided in four parts. In the first, I describe the datasets used to construct our database. In the second, the construction of the sample of scientists is presented with an emphasis on network definition. In the third part, I document the construction of the variables. Finally, I provide descriptive statistics.

4.1. Data Sources

Publication records. My primary data source is the MEDLINE publication database.¹ MEDLINE, the US National Library of Medicine’s primary bibliographic database, includes almost all journal articles in life sciences dating back to 1946. It provides a wide range of data elements on journal articles, such as author names, publication dates, journal titles, grant acknowledgments, and Medical Subject Headings (MeSH) terms.

MeSH terms are a controlled vocabulary maintained by the National Library of Medicine, offering a detailed categorization of the intellectual space covered by the biomedical research literature. For our purposes, it is important to note that MeSH terms are assigned to each publication by professional indexers who focus exclusively on the scientific content of the papers.² This ensures that MeSH terms

¹I restrict the dataset to research articles, excluding publication types such as book reviews, letters, and other non-research outputs.

²The Medical Subject Headings Section staff subject specialists are responsible for areas of the health sciences in which they have knowledge and expertise. Specialists from various disciplines are also consulted regarding organizational changes, and close coordination is maintained with specialized vocabularies.

provide an objective description of the paper’s scientific content.

To obtain citation data for the articles, I use the NIH Open Citation Collection (Hutchins et al., 2019). To focus on the relevant research within oncology, I extracted all articles with at least one relevant MeSH term. Relevant MeSH terms are defined as any child term of ”Neoplasms” in the MeSH tree.

Author name disambiguation. Author names alone are not reliable identifiers for individuals, as multiple authors can share the same name, and authors’ names and affiliations may change over time. Author name disambiguation is a longstanding challenge in gathering author-specific publication data, particularly because bibliographic databases typically lack unique individual author identifiers. In the case of MEDLINE, nearly two-thirds of authors face name ambiguity, where their last name and first initial are shared by one or more other authors (Smalheiser et al., 2009). The so-called ”John Smith” problem has been the focus of numerous disambiguation algorithms that attempt to identify individuals based on their author names.³

Despite efforts to create global author identifiers, such as ORCID and ResearcherID, many articles in MEDLINE, especially those published before 2003 (when the ORCID field was added to PubMed), provide limited author information, typically only the last name, first initial, and, for first authors before 2014, the affiliation.

I rely on the PubMed Knowledge Graph (PKG) introduced by Xu et al. (2020) to tackle the issue of ambiguous author names. Their algorithm computes clusters of articles that are likely authored by the same individual. This method has shown outstanding accuracy, especially for NIH-funded scientists. Their approach builds on the probabilistic model developed by Torvik et al. (2005), which assumes that articles written by the same author tend to share more common attributes than articles written by different authors.

Author Attributes. To complement the two primary datasets, I incorporate author-level variables derived from metadata compiled by the Torvik research group and its associated projects. Specifically, I focus on authors’ ethnicity, gender, and affiliation.

The *Ethnea* tool is used to determine authors’ ethnicity based on their first and last names.⁴ Unlike other ethnicity matching algorithms, *Ethnea* primarily reflects an author’s nationality, rather

³Author name disambiguation remains one of the major unresolved issues in bibliometrics. Kang et al. (2009) provide an excellent review of the problem and the various approaches developed to address it.

⁴The ”Ethnea” tool, developed by the University of Illinois, assigns ethnicity based on name matching. It is available at <http://abel.lis.illinois.edu/cgi-bin/ethnea/search.py>.

than distant ancestral origins, and it also accounts for dual ethnicities that may arise from factors like migration or marriage.

Gender information is assigned through the *Genni* tool, which uses a probabilistic model to match first names to gender. This model also incorporates the ethnicity associated with the last name. For example, the name "Andrea" is identified as female if the associated ethnicity is French, but male if the ethnicity is Italian.

Authors' affiliations are obtained using the *MapAffil* tool, which links PubMed authors' affiliation data to cities and geocodes worldwide. The MapAffil 2016 dataset, based on a snapshot of PubMed from October 2016, connects affiliation strings to specific authors and articles. While PubMed recorded only the first author's affiliation prior to 2014, the MapAffil 2016 dataset extends coverage to include records lacking affiliation information, sourced from databases such as PMC, NIH grants, the Microsoft Academic Graph, and the Astrophysics Data System. It is important to note that the dataset does not include data after 2015 and covers only 62.9% of affiliation instances in PubMed.

In addition to ethnicity, gender, and affiliation, I also record each author's degree(s), the year of degree completion, and whether they have been recognized as prize-winning scientists. This comprehensive set of author attributes enables the construction of detailed profiles, helping to control for demographic, educational, and professional characteristics that influence scientific productivity. Specifically, the inclusion of degree information provides insight into an author's academic background and level of experience, while prize recognitions help identify individuals with notable accomplishments in their field, allowing for control of heterogeneity in productivity.

4.2. Network Definition

My analysis focuses on authors within the field of cancer research, where collaboration is a common practice and publication output serves as a reliable metric for productivity. Scientific knowledge, particularly in its early stages, is often tacit and confined to small, close-knit groups. As such, collaborations play a crucial role in the exchange of information and the generation of new ideas (Nonaka and Takeuchi, 2007).

To ensure that the sample accurately represents active researchers rather than practitioners with sporadic or incidental publications, I apply specific inclusion criteria. Specifically, I restrict the sample to scientists who graduated before the beginning of the period (the earlier of their PhD or MD completion date for those with dual degrees). Additionally, I require each author to have at least one publication (with a citation) as either the first or last author in the cancer research field. This practice aligns

with a well-established social norm in the life sciences, where first authorship is assigned to the junior researcher responsible for conducting the work, and last authorship is reserved for the principal investigator leading the study. Middle authors receive credit based on their position in the author list, with recognition generally diminishing as one moves further from either end (Zuckerman, 1968; Nagaoka and Owan, 2014).

In the context of the co-authorship network, two authors are considered linked if they co-authored at least one paper over the period in the field of cancer. This results in a network with a large central component and several smaller, disconnected components.

To identify tightly-knit groups within the largest connected component of the network, I use a modularity-based community detection algorithm developed by Traag et al. (2019). This algorithm identifies a partition of the network that approximately maximizes the modularity score, which measures how strongly the network is organized into communities.⁵ For the purposes of this analysis, I include only those communities with at least 10 authors. I pooled all the communities in the same network, so that each community forms a component of the considered network.

4.3. Variable Definitions

IV Construction: To measure homophily among co-authors, I adopt the framework proposed by Boschma (2005), which identifies five key dimensions of proximity: cognitive, social, institutional, geographical, and cultural. Due to the high correlation between geographical and institutional proximity in the dataset, the geographical dimension is excluded from the analysis.

- **Cognitive proximity** refers to the shared knowledge and experience among authors within a field, emphasizing domain-specific expertise rather than research methods or practices (Nooteboom, 2000). Research overlap, which captures the similarity of research interests, is a key factor influencing collaboration. This is consistent with studies like Fafchamps et al. (2010), which find that co-authorship is more likely when researchers share common research interests.

To measure cognitive proximity, I link cancer research articles published within the first three years of a researcher’s career to second-level MeSH (Medical Subject Headings) terms. Using these MeSH profiles, I compute cosine similarity between authors’ research interests to quantify their cognitive proximity (Breschi et al., 2003).

⁵Modularity-based algorithms are widely studied in fields like computer science and physics (e.g., Loscalzo and Barabási (2016), Ch. 9).

- **Social proximity** refers to relationships between scientists, often developed through prior collaborations, that help reduce uncertainty about the potential productivity of peers (Uzzi, 1996). Breschi and Lissoni (2009) highlight that social ties, captured through past co-authorships, play a significant role in increasing the likelihood of future collaborations. In this study, I define social proximity through the structure of the co-authorship network over the period from $t - 5$ and $t - 1$, considering both direct collaborations and indirect connections through shared co-authors.
- **Institutional proximity** emphasizes the importance of face-to-face interactions and shared institutional resources in fostering scientific collaborations. Studies show that researchers working within the same department or institution are more likely to collaborate (Dahlander and McFarland, 2013; Kabo et al., 2014; Long et al., 2014). To measure institutional proximity, I used MapAffil tools to identify authors’ affiliations, designating their primary affiliation during a specific period as the institution where they published the most.
- **Cultural proximity** reflects the influence of shared social norms, language, or cultural backgrounds in promoting collaboration. I used authors’ ethnicity as a proxy for cultural proximity, utilizing the Ethnea tool.⁶ This aligns with previous studies, such as Dahlander and McFarland (2013) Freeman and Huang (2015), which found that ethnic similarity increases the likelihood of co-authorship.

Alongside these proximity dimensions, gender is considered a key factor influencing collaboration. Gender can affect co-authorship dynamics in several ways. First, the homophily argument suggests that women may prefer to collaborate with other women, while men may show a similar preference for male co-authors. Second, gender may introduce a potential discrimination effect, where male co-authors are favored over female ones. Finally, gender may influence collaboration decisions due to risk aversion, as discussed by Ductor et al. (2023), which could shape the selection process for projects perceived as risky.

Author Productivity. Demographic information for each scientist is primarily obtained from curriculum vitae (CVs), which provide details on academic degrees (e.g., bachelor’s, MD, PhD, or MD/PhD), including the dates of completion and the institutions where they were earned. In addition, gender is

⁶Ethnea maps people into 26 ethnicity groups, including English, Hispanic, Chinese, German, Japanese, French, Italian, Indian, Arab, and others.

inferred using metadata from the Torvik Research Group.

I also compute each author’s hit, defined as the article with the highest cumulative citation count among those where the author appears as either the first or last author.

For late-career recognition, I include individuals elected as Members of the National Academy of Sciences or the Institute of Medicine before the start of the period. I also account for current and former Howard Hughes Medical Investigators (HHMIs), a distinction awarded every three years by the Howard Hughes Medical Institute to select mid-career biomedical scientists based on their potential for significant contributions to their fields.

Table 4.1: Summary Statistics for the period 2013 2015.

	N	Mean	St. Dev.	Min	Max
Asinh(Production Period)	1,392	5.9	1.6	0.9	11.1
Asinh(hit)	1,392	4.6	1.2	0.0	8.3
Asinh(Decades after Undergraduate)	1,392	1.7	0.4	0.7	2.8
Phd	1,392	0.5	0.5	0	1
Late Career Accolade	1,392	0.1	0.3	0	1
Male	1,392	0.7	0.5	0	1

5. Empirical findings

5.1. First step: collaboration choices

I start my empirical analysis by assessing the effectiveness of the network formation model employed to instrument the observed network. Table (5.1) presents the estimation results for Model (5). I find that co-authorship networks are influenced by social proximity through prior collaboration networks. Specifically, being a coauthor or sharing a common coauthor in the previous period’s network significantly increases the likelihood of collaboration. This supports the notion that social proximity increases the likelihood of collaboration, in line with earlier studies highlighting its importance. A plausible explanation is that network proximity gives a better access to information regarding the skills and productivity of potential coauthors.

Researchers at the same institution are more likely to collaborate, given easier access to shared resources such as laboratories, hospitals, and advanced technologies (e.g., genome sequencing). Informal interactions among researchers in close proximity may also promote collaboration.

Cultural proximity also significantly influences collaboration, with shared ethnicity emerging as a key

predictor of collaboration intensity. This finding is consistent with the broader homophily literature, suggesting that individuals from similar cultural backgrounds may incur lower communication and coordination costs, thus facilitating more efficient collaboration. Interestingly, this effect persists even among highly productive scientists, which contrasts with Freeman and Huang (2015), who found no ethnic homophily among leading economists.

Finally, the relatively high McFadden Pseudo R^2 observed in the model indicates that the weak

Table 5.1: Logistic regression of link formation for the period 2013-2015.

Cosine Similarity	1.317*** (0.199)
Same Grad school	0.484** (0.214)
Same affiliation	0.988*** (0.046)
Past coauthor	2.757*** (0.062)
Past common coauthor	1.583*** (0.051)
Same Ethnicity	0.170*** (0.045)
Same gender	0.058 (0.046)
Observations	55,952
Log Likelihood	-8,053.078
Akaike Inf. Crit.	16,122.160
Mc Fadden Pseudo R^2	0.3011

Note Results for Model (5) of the article are displayed. The dependent variable is defined as the existence of collaboration between author i and author j . The independent variables capture differences in characteristics between i and j . A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

instrument problem is unlikely.⁷

5.2. On the root of peer effects

Table 5.2 reports the estimation results for the general econometric model (3) using alternative estimators that consider whether the adjacency matrix is endogenous or not in their construction. The

⁷In the case of a weak-IV problem, the GMM approach developed by Liu et al. (2014) can be employed, although some adjustments are necessary to account for heteroscedasticity.

Table 5.2: Estimation results for the period 2013–2015.

	<i>Dependent variable:</i> Weighted number of citations					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
ϕ_1	0.015*** (0.001739)	0.016*** (0.002455)			0.015*** (0.001737)	0.016*** (0.002431)
ϕ_2	0.128* (0.058001)	0.287** (0.104428)	0.11 (0.059356)	0.354** (0.109494)		
<i>Asinh</i> (Hit)	0.52*** (0.04281)	0.503*** (0.044707)	0.596*** (0.044341)	0.574*** (0.046363)	0.533*** (0.041733)	0.528*** (0.041829)
<i>Asinh</i> (Decades after undergraduate)	-0.688*** (0.112219)	-0.659*** (0.116197)	-0.699*** (0.114932)	-0.655*** (0.120506)	-0.712*** (0.111389)	-0.711*** (0.111434)
Phd	0.324*** (0.082158)	0.295*** (0.085362)	0.257** (0.084095)	0.208* (0.087669)	0.349*** (0.080225)	0.353*** (0.080772)
Late Career Accolade	0.422** (0.159383)	0.417* (0.162776)	0.5** (0.160086)	0.497** (0.167051)	0.424** (0.157447)	0.42** (0.156854)
Male	0.139 (0.076626)	0.112 (0.079636)	0.137 (0.078896)	0.095 (0.082645)	0.161* (0.076338)	0.161* (0.076368)
Cragg-Dold Wald F statistic	25.9847	22.2752	27.6621	26.798	33.73	28.6599
OIR test p-value	0.0081	0.432	0.0063	0.759	7e-04	0.306

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$, as instruments for the observed adjacency matrix $\mathbf{G}$. Columns 1 and 2 report estimates for the mixed model, columns 3 and 4 for the local-average model, and the last two columns for the local aggregate model. The dependent variable is the number of citations over the study period, weighted by the journal impact factor. The following dummy variables are included: Male, PhD, and Late Career Recognition. Experience is captured by the variable Decades after undergraduate studies. Hit is computed using the inverse hyperbolic sine (*asinh*) of the maximum citations received for an article. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by *, **, and ***, which denote significance at the 10%, 5%, and 1% levels.

2SLS-1 estimates are only valid if the network adjacency matrix $\mathbf{G}$ is exogenous conditional on control variables $\mathbf{X}$ and network fixed effects. In the case of the 2SLS-2 estimator, the p-values provide strong evidence that my IV strategy successfully corrects for the bias arising from the endogeneity of $\mathbf{G}$.

Which mechanisms are driving scientific peer effects? What is the impact of neglecting one significant mechanism on the assessment of the second? The first question can be answered by the statistical significance of ϕ_1 and ϕ_2 in the general network model. My results indicate that scientific peer effects find their roots in a combination of strategic complementarity and conformity. However, estimating either model in isolation may result in misleading conclusions regarding the magnitude of the effect.

My findings highlight important caveats in the empirical analysis of scientific peer effects. Peer effects are a complex phenomenon, and their assessment should be approached with caution. If more than one mechanism is driving scientific peer effects, neglecting one of them can produce biased inferential results.

I also report the empirical results obtained when separately estimating the local-aggregate and the local-average models. Comparing columns 1–6, it is evident that the local-average peer effect is overstated when the local-aggregate effect is not accounted for, while, surprisingly, the estimate of strategic

complementarity remains unchanged. As evidence by estimates reported in the appendix, these results are consistent accross period.

6. Concluding remarks

This paper contributes to the literature on scientific peer effects by providing microfoundations for how collaboration influences research productivity. Focusing on coauthorship in cancer research, I demonstrate that peer effects can be credibly identified outside of natural experiments by employing an instrumental variables strategy that addresses network endogeneity. The analysis highlights the importance of distinguishing among underlying mechanisms, particularly strategic complementarity and conformity, in shaping collaborative behavior and knowledge production. These findings offer new insight into the process of scientific discovery and carry implications for the effective organization of research teams and the design of science policy.

Let me mention a few possible extensions worth pursuing. First, while I model conformity as deviation from the average action of peers, an alternative would be to focus on salient individuals, such as high achievers or low performers. Second, it would be interesting to explore heterogeneity in peer effects not only through the network structure but also as a function of individual characteristics. Third, one could examine the interplay between coauthorship and inventorship networks, particularly in biomedical fields where researchers often participate in both. This could shed light on the strategic relationship between scientific and technological production. I leave these exciting topics for future research.

Appendix A. Robustness check

Table A.1: Summary Statistics for the period 2010–2012.

Statistic	N	Mean	St. Dev.	Min	Max
Asinh(Production Period)	1,078	5.9	1.6	0.9	10.7
Asinh(hit)	1,078	4.5	1.2	0.0	8.0
Asinh(Decades after Undergraduate)	1,078	1.7	0.3	0.7	2.6
Phd	1,078	0.4	0.5	0	1
Late Career Accolade	1,078	0.1	0.2	0	1
Male	1,078	0.7	0.4	0	1

Table A.2: Logistic Regression of Link Formation (2010–2012)

Cosine Similarity	1.589*** (0.233)
Same Grad School	0.423 (0.264)
Same Affiliation	0.782*** (0.056)
Past Coauthor	2.598*** (0.070)
Past Common Coauthor	1.468*** (0.063)
Same Ethnicity	0.053 (0.054)
Same Gender	0.211*** (0.056)
Constant	−4.716*** (0.083)
Observations	37,864
Log Likelihood	−5,653.252
Akaike Inf. Crit.	11,322.500
McFadden R^2	0.2818

Note Results for Model (5) of the article are displayed. The dependent variable is defined as the existence of collaboration between author i and author j . The independent variables capture differences in characteristics between i and j . A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table A.3: Estimation results for the period 2010–2012.

	<i>Dependent variable:</i>					
	Weighted number of citations					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
ϕ_1	0.022*** (0.002057)	0.023*** (0.003149)			0.021*** (0.002065)	0.024*** (0.003186)
ϕ_2	0.168* (0.070267)	0.328* (0.143987)	0.148* (0.072938)	0.55*** (0.166632)		
<i>Asinh</i> (Hit)	0.447*** (0.043903)	0.429*** (0.046612)	0.525*** (0.045791)	0.491*** (0.048964)	0.463*** (0.043585)	0.453*** (0.044484)
<i>Asinh</i> (Decades after undergraduate)	-0.507*** (0.132012)	-0.477*** (0.138829)	-0.471*** (0.138497)	-0.389* (0.153946)	-0.54*** (0.130604)	-0.545*** (0.130779)
Phd	0.214* (0.090349)	0.183 (0.097251)	0.092 (0.092584)	-0.001 (0.101569)	0.25** (0.089152)	0.266** (0.091131)
Late Career	0.323 (0.181707)	0.267 (0.19157)	0.601** (0.192148)	0.498* (0.20455)	0.372* (0.180162)	0.336 (0.185333)
Male	0 (0.092811)	-0.003 (0.094241)	0.052 (0.097451)	0.05 (0.101594)	0.003 (0.093585)	-0.004 (0.093479)
Cragg-Dold Wald F statistic	17.6138	14.9592	16.5226	17.1218	23.5695	19.6358
OIR test p-value	0.6311	0.4045	0.5176	0.5915	0.0067	0.168

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$, as instruments for the observed adjacency matrix $\mathbf{G}$. Columns 1 and 2 report estimates for the mixed model, columns 3 and 4 for the local-average model, and the last two columns for the local aggregate model. The dependent variable is the number of citations over the study period, weighted by the journal impact factor. The following dummy variables are included: Male, PhD, and Late Career Recognition. Experience is captured by the variable Decades after undergraduate studies. Hit is computed using the inverse hyperbolic sine (*asinh*) of the maximum citations received for an article. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by *, **, and ***, which denote significance at the 10%, 5%, and 1% levels.

Table A.4: Summary Statistics for the period 2011–2013

Statistic	N	Mean	St. Dev.	Min	Max
<i>Asinh</i> (Production Period)	1,217	5.9	1.5	0.9	10.7
<i>Asinh</i> (hit)	1,217	4.6	1.2	0.0	8.0
<i>Asinh</i> (Decades after UG)	1,217	1.7	0.3	0.6	2.8
PhD	1,217	0.4	0.5	0	1
Late Career Accolade	1,217	0.1	0.3	0	1
Male	1,217	0.7	0.5	0	1

Table A.5: Logistic Regression of Link Formation (2011–2013)

Cosine Similarity	1.886***
	(0.221)
Same Grad School	0.119
	(0.286)
Same Affiliation	0.887***
	(0.053)
Past Coauthor	2.713***
	(0.069)
Past Common Coauthor	1.496***
	(0.059)
Same Ethnicity	0.009
	(0.050)
Same Gender	−0.015
	(0.052)
Observations	45,925
Log Likelihood	−6,458.510
AIC	12,933.020
McFadden R^2	0.2972

Note Results for Model (5) of the article are displayed. The dependent variable is defined as the existence of collaboration between author i and author j . The independent variables capture differences in characteristics between i and j . A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table A.6: Estimation Results for the Period 2011–2013

	<i>Weighted Number of Citations</i>					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
ϕ_1	0.018*** (0.002)	0.023*** (0.003)			0.018*** (0.002)	0.023*** (0.003)
ϕ_2	0.166* (0.070)	0.302* (0.143)	0.155* (0.071)	0.557*** (0.147)		
Asinh(Hit)	0.435*** (0.043)	0.407*** (0.046)	0.511*** (0.044)	0.482*** (0.047)	0.448*** (0.043)	0.427*** (0.044)
Asinh(Decades after UG)	-0.441*** (0.120)	-0.418*** (0.123)	-0.449*** (0.126)	-0.386** (0.135)	-0.467*** (0.122)	-0.465*** (0.122)
PhD	0.254** (0.086)	0.253** (0.096)	0.146 (0.088)	0.060 (0.098)	0.287*** (0.084)	0.318*** (0.087)
Late Career	0.313 (0.180)	0.279 (0.186)	0.427* (0.184)	0.411* (0.197)	0.322 (0.178)	0.290 (0.179)
Male	0.163 (0.086)	0.166 (0.088)	0.153 (0.088)	0.156 (0.094)	0.162 (0.086)	0.164 (0.086)
F-statistic	19.15	17.46	18.79	19.54	23.96	22.50
OIR p-value	0.060	0.463	0.614	0.902	0.253	0.321

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$, as instruments for the observed adjacency matrix $\mathbf{G}$. Columns 1 and 2 report estimates for the mixed model, columns 3 and 4 for the local-average model, and the last two columns for the local aggregate model. The dependent variable is the number of citations over the study period, weighted by the journal impact factor. The following dummy variables are included: Male, PhD, and Late Career Recognition. Experience is captured by the variable Decades after undergraduate studies. Hit is computed using the inverse hyperbolic sine (asinh) of the maximum citations received for an article. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by *, **, and ***, which denote significance at the 10%, 5%, and 1% levels.

Table A.7: Summary Statistics for the period 2012–2014.

Statistic	N	Mean	St. Dev.	Min	Max
Asinh(Production Period)	1,316	5.9	1.5	0.9	10.7
Asinh(hit)	1,316	4.6	1.2	0.0	8.3
Asinh(Decades after Undergraduate)	1,316	1.7	0.4	0.7	2.8
Phd	1,316	0.5	0.5	0	1
Late Career Accolade	1,316	0.1	0.2	0	1
Male	1,316	0.7	0.5	0	1

Table A.8: Logistic regression of link formation for the period 2012–2014.

Cosine Similarity	1.378*** (0.207)
Same Grad school	0.062 (0.260)
Same affiliation	0.929*** (0.050)
Past coauthor	2.715*** (0.066)
Past common coauthor	1.554*** (0.055)
Same Ethnicity	0.153*** (0.048)
Same gender	0.142*** (0.049)
Observations	50,397
Log Likelihood	−7,236.084
Akaike Inf. Crit.	14,488.170
Mc Fadden Pseudo R^2	0.2949

Note Results for Model (5) of the article are displayed. The dependent variable is defined as the existence of collaboration between author i and author j . The independent variables capture differences in characteristics between i and j . A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table A.9: Estimation results for the period 2012 2014.

	<i>Dependent variable:</i>					
	Weighted number of citations					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
ϕ_1	0.018*** (0.001677)	0.018*** (0.002534)			0.018*** (0.0017)	0.02*** (0.002474)
ϕ_2	0.259*** (0.05979)	0.363*** (0.096034)	0.279*** (0.062987)	0.525*** (0.102846)		
<i>Asinh</i> (Hit)	0.413*** (0.041123)	0.398*** (0.043268)	0.493*** (0.04252)	0.466*** (0.044292)	0.44*** (0.040367)	0.433*** (0.041679)
<i>Asinh</i> (Decades after undergraduate)	-0.566*** (0.117828)	-0.55*** (0.119669)	-0.606*** (0.12178)	-0.572*** (0.127245)	-0.601*** (0.117923)	-0.597*** (0.118041)
Phd	0.208* (0.083193)	0.198* (0.087632)	0.091 (0.084761)	0.058 (0.088434)	0.245** (0.082025)	0.255** (0.083921)
<i>late_career</i>	0.607*** (0.16445)	0.605*** (0.167999)	0.708*** (0.167185)	0.711*** (0.177296)	0.602*** (0.159567)	0.594*** (0.160084)
Male	0.113 (0.080001)	0.106 (0.082009)	0.107 (0.082484)	0.09 (0.086308)	0.13 (0.079808)	0.131 (0.079858)
Cragg-Dold Wald F statistic	22.9799	19.6143	23.2529	23.6743	29.4584	24.6042
OIR test p-value	0.5448	0.6734	0.2109	0.3933	0.0019	0.1562

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$, as instruments for the observed adjacency matrix $\mathbf{G}$. Columns 1 and 2 report estimates for the mixed model, columns 3 and 4 for the local-average model, and the last two columns for the local aggregate model. The dependent variable is the number of citations over the study period, weighted by the journal impact factor. The following dummy variables are included: Male, PhD, and Late Career Recognition. Experience is captured by the variable Decades after undergraduate studies. Hit is computed using the inverse hyperbolic sine (*asinh*) of the maximum citations received for an article. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by *, **, and ***, which denote significance at the 10%, 5%, and 1% levels.

Table A.10: Summary Statistics for the period 2014 2016.

	N	Mean	St. Dev.	Min	Max
<i>Asinh</i> (Production Period)	1,539	5.8	1.6	1.0	10.9
<i>Asinh</i> (hit)	1,539	4.6	1.2	0.0	8.3
<i>Asinh</i> (Decades after Undergraduate)	1,539	1.7	0.4	0.7	2.5
Phd	1,539	0.5	0.5	0	1
Late Career Accolade	1,539	0.1	0.3	0	1
Male	1,539	0.7	0.5	0	1

Table A.11: Logistic Regression of Link Formation (2014–2016)

Cosine Similarity	1.201*** (0.181)
Same Grad School	0.657*** (0.218)
Same Affiliation	1.076*** (0.043)
Past Coauthor	2.827*** (0.053)
Past Common Coauthor	1.592*** (0.049)
Same Ethnicity	0.156*** (0.042)
Same Gender	0.106** (0.042)
Constant	−5.008*** (0.063)
Observations	80,974
Log Likelihood	−9,748.737
Akaike Inf. Crit.	19,513.470
McFadden R^2	0.3225

Note Results for Model (5) of the article are displayed. The dependent variable is defined as the existence of collaboration between author i and author j . The independent variables capture differences in characteristics between i and j . A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Table A.12: Estimation Results for the Period 2014–2016

<i>Dependent variable:</i>						
Weighted number of citations						
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
ϕ_1	0.013*** (0.001549)	0.013*** (0.001986)			0.013*** (0.001556)	0.013*** (0.001913)
ϕ_2	0.138** (0.053458)	0.312** (0.101474)	0.14* (0.056405)	0.383*** (0.110943)		
<i>Asinh</i> (Hit)	0.555*** (0.039683)	0.534*** (0.042009)	0.642*** (0.040011)	0.616*** (0.042249)	0.572*** (0.039042)	0.568*** (0.03944)
<i>Asinh</i> (Decades after undergraduate)	-0.837*** (0.105211)	-0.774*** (0.112121)	-0.846*** (0.108644)	-0.759*** (0.117689)	-0.887*** (0.103435)	-0.887*** (0.103455)
Phd	0.133 (0.073547)	0.102 (0.077632)	0.043 (0.074683)	-0.002 (0.078469)	0.157* (0.073005)	0.161* (0.074115)
<i>late_career</i>	0.531*** (0.140627)	0.539*** (0.143711)	0.563*** (0.140861)	0.576*** (0.14492)	0.524*** (0.140238)	0.523*** (0.140772)
Male	0.122 (0.0722)	0.112 (0.073737)	0.153* (0.074512)	0.141 (0.076826)	0.129 (0.072068)	0.128 (0.072088)
Cragg-Dold Wald F statistic	28.5238	25.6933	31.2342	32.2909	38.0249	33.034
OIR test p-value	0.0284	0.1464	0.4066	0.0948	2e-04	0.0988

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$, as instruments for the observed adjacency matrix $\mathbf{G}$. Columns 1 and 2 report estimates for the mixed model, columns 3 and 4 for the local-average model, and the last two columns for the local aggregate model. The dependent variable is the number of citations over the study period, weighted by the journal impact factor. The following dummy variables are included: Male, PhD, and Late Career Recognition. Experience is captured by the variable Decades after undergraduate studies. Hit is computed using the inverse hyperbolic sine (*asinh*) of the maximum citations received for an article. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by *, **, and ***, which denote significance at the 10%, 5%, and 1% levels.

Appendix B. Cancer related papers

To identify cancer-relevant papers, we leverage MeSH (Medical Subject Headings) terms, a hierarchical controlled vocabulary thesaurus managed by the National Library of Medicine (NLM). Professional indexers at the NLM assign MeSH terms to biomedical publications following established protocols, ensuring consistency and context within the entire collection of articles. Significantly, authors of the publications do not participate in selecting MeSH terms. This ensures that the indexing process remains objective, as it is carried out by trained professionals who mitigate the inherent subjectivity of indexing tasks.

MeSH terms serve as a critical tool in biomedical research, enabling researchers to effectively search and categorize vast amounts of literature. By organizing articles into a structured hierarchy, MeSH terms facilitate precise and efficient retrieval of information, which is crucial for advancing scientific knowledge and improving clinical practice. The hierarchical structure of MeSH terms allows for the grouping of related concepts, making it easier to identify overarching themes and specific details within the literature.

The figure below illustrates the MeSH tree structure for cancer-related terms, showcasing how

specific terms are nested under broader categories. This visual representation helps in understanding the relationships and hierarchies among different MeSH terms, highlighting their interconnectedness and relevance to cancer research.

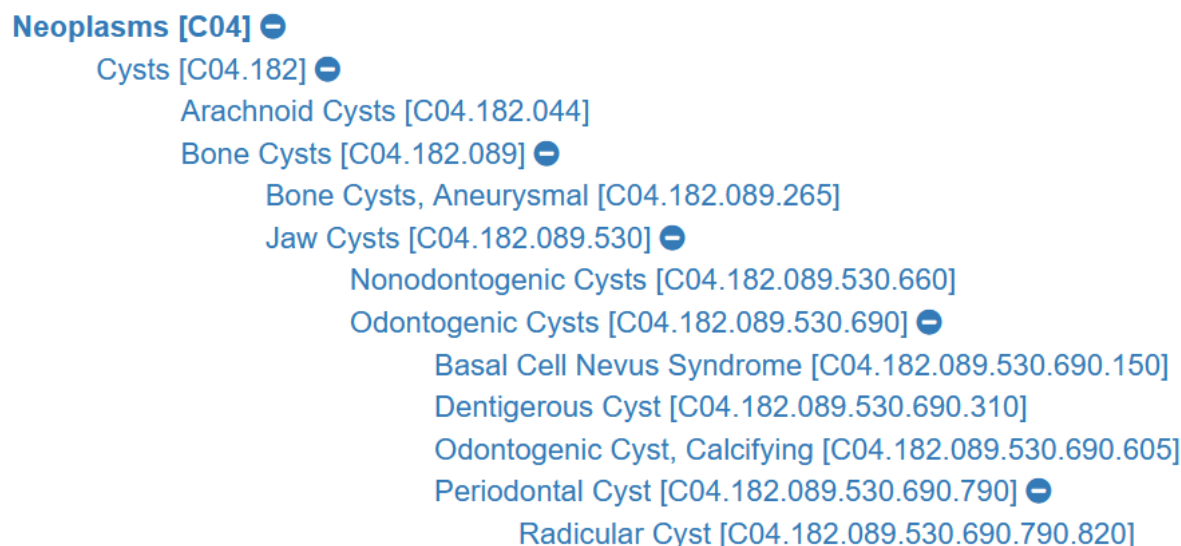


Figure B.1: Illustration of the MeSH tree structure for Neoplasms child nodes.

Cancer-related papers were identified primarily using the child of the MeSH code C04, i.e., "Neoplasms," which represents the most comprehensive family of MeSH terms for cancer. This category encompasses a wide range of malignancies, including solid tumors and hematologic cancers, providing a broad foundation for cancer research. In addition to C04, we considered the advice of actual oncologists to include children of the following MeSH terms, which are relevant to various aspects of cancer biology, treatment, and research:

- **Drug Screening Assays, Antitumor** (E05.337.550.200): These assays are crucial for identifying and developing new antitumor agents by testing their efficacy in inhibiting cancer cell growth.
- **Cancer Vaccines** (D20.215.894.200): Vaccines designed to prevent or treat cancer by stimulating the immune system to target cancer cells.
- **Neoplasms** (C04): A broad category encompassing all types of tumors, both benign and malignant.
- **DNA, Neoplasm** (D13.444.308.425): Refers to the genetic material specific to

cancer cells, which can be targeted for diagnosis or therapy.

- **Drug Resistance, Neoplasm** (G07.690.773.984.395): The ability of cancer cells to resist the effects of chemotherapy, posing a significant challenge in cancer treatment.
- **Neoplasm Proteins** (D12.776.624): Proteins specifically associated with tumors, which can serve as biomarkers or therapeutic targets.
- **Biomarkers, Tumor** (D23.101.140): Biological molecules found in blood, other body fluids, or tissues that indicate the presence of cancer.
- **Antigens, Neoplasm** (D23.050.285): Substances produced by tumor cells that can trigger an immune response.
- **Oncogenic Viruses** (B04.613): Viruses that can cause cancer by integrating their genetic material into the host genome.
- **Tumor Cells, Cultured** (A11.251.860): Cancer cells grown in laboratory conditions for research purposes.
- **Neoplasm Proteins** (D12.776.624): Proteins associated with tumors, important for understanding cancer biology and developing treatments.
- **Chemotherapy, Cancer, Regional Perfusion** (E04.292.425): A technique to deliver high doses of chemotherapy directly to the tumor site.
- **Antineoplastic Agents** (D27.505.954.248): Drugs used to treat cancer by inhibiting the growth of malignant cells.
- **Receptors, Tumor Necrosis Factor** (D12.776.543.750.705.852.760): Receptors involved in the signaling pathways that can lead to tumor cell death.
- **Tumor Escape** (G12.900): Mechanisms by which cancer cells evade the immune system.
- **Neoplastic Stem Cells** (A11.872.650): Stem cells within tumors that have the ability to self-renew and drive cancer progression.
- **Carcinogens** (D27.888.569.100): Substances capable of causing cancer in living tissue.
- **Gammaretrovirus** (B04.820.650.375): A type of virus that can insert its genetic material into the host genome, potentially causing cancer.
- **Antibodies, Neoplasm** (D12.776.377.715.548.114.240): Antibodies used to target and neutralize cancer cells.

- **Receptors, Immunologic** (D12.776.543.750.705): cer cells.

Receptors on immune cells that can be manipulated to enhance the immune response against cancer.

- **Biomarkers, Tumor** (D23.101.140): Indicators used to detect cancer or monitor its progression.

- **Tumor Necrosis Factors** (D23.529.374.750): Proteins involved in the destruction of can-

- **Radiotherapy** (E02.815): The use of high-energy radiation to kill or shrink cancer cells.

Appendix C. Computing Cognitive proximity

To measure the distance, or rather proximity, in intellectual or "ideas space" between pairs of scientists, we constructed a variable based on MeSH terms. Delineating the boundaries of scientific fields is challenging, as most research can be classified in numerous ways, and consensus among scientists on specific categorizations is often lacking. Traditional measures based on shared department affiliation or broad scientific field distinctions (e.g., cell vs. molecular biology) are inadequate for this purpose. Instead of positioning scientists at fixed points in ideas space, our method provides a cost-effective and convenient way to measure their relative positions.

MeSH terms offer a fine-grained level of detail, making them particularly useful for this task. For example, a paper motivating the 2018 Nobel Prize in Medicine (Ishida et al., 1992) is labeled with 27 distinct descriptors, ranging from general terms like "Animals" and "Humans" to specific ones like "CD3 Complex/Genetics" and "T Cells/Immunology." However, during the construction of MeSH vectors for 2001-2005, we found that such detailed categories could result in overly specific and uninformative comparisons across scientists. Fine-grained categorizations might obscure meaningful patterns by focusing too narrowly on minor differences.

To ensure more informative and comparative analysis, we opted for a coarser categorization of MeSH terms, utilizing second-level categories. For instance, the MeSH term "Receptors, Immunologic," originally categorized under the tree number D12.776.543.750.705, was grouped under the broader category D12.776. We then constructed a vector for each author, where each entry is set to 1 if the author has at least one paper with a MeSH term belonging to the specified categories over the period, and 0 otherwise. This approach allowed us to capture broader trends and more meaningful comparisons between scientists' intellectual proximity.

The cosine index (Breschi et al., 2003) was used to calculate the extent of overlap between the author’s research interest and each of their co-authors’ interests. Here, it was assumed that the higher the overlap between two authors in terms of the breadth and depth of their research interest, the closer they are in the cognitive space. The cosine index between author i and j , which is used as the independent variable $CognProx_{ij}$, is calculated as follows:

$$CognProx_{ij} = \frac{\sum_k a_{ik}a_{jk}}{\sqrt{\sum_k a_{ik}^2}\sqrt{\sum_k a_{jk}^2}}$$

where a_{ik} refers to the presence of categorie k mesh in all the papers published by author i . Obviously, $CognProx_{ij} = 1$ indicates that the two authors are exactly the same in terms of their research interest, and if there is no common research interest between the two authors, $CognProx_{ij} = 0$. Therefore, high cosine values indicate increased overlap between the knowledge bases of two authors, in terms of their similarity.

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